Launch vehicle overview

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1 Introduction

One aspect that differentiates CubeSats from conventional spacecraft is the large availability of launch opportunities. This is due to both the interface standardization and containerization [1]. Since the first CubeSat launches in the early 2000s, these satellites have employed the shared payload approach [2] with more than 1000 launched during the following two decades [3]. Only recently have CubeSat clusters become the primary payloads of commercial launch vehicles thanks to the introduction of micro launch vehicles, such as Rocket Labs' Electron [2]. CubeSats have become a frequent element of low Earth orbit launches, although they still compose a small part of the total mass launched to space. Even though PSLV-C37 [4] launched 101 CubeSats, the combined mass of those satellites weighed less than its main payload CARTOSAT 2D (Fig. 1).

Despite the large number of CubeSats launched, the overall participation of nanosatellites in the space launch economy remains small: small satellites (under 500 kg) [5] represented <2%, and CubeSats <0.2% of the overall mass launched between 2013 and 2017. Regardless of that, CubeSats still form a distinctive niche with defined characteristics. The importance of micro payloads (<100 kg), of which CubeSats are the majority both in number and total launch mass, has led to the development of several micro launch vehicles, such as Electron, CZ-11, and Kuaizhou.

The advent of small commercial launch vehicles may well have a deep impact on the design of CubeSat missions, since dedicated launches will allow mission planners to have a considerably greater freedom of design. The participation and importance of micro launch vehicles is expected to increase: until 2017 <1% of all CubeSats were launched on micro launchers; between 2018 and 2019 12% [2] of all CubeSats were flown on micro launch vehicles.

The CubeSat landscape is dominated by the Planet Labs and Spire constellation launches, and this characteristic is reflected in the launch vehicles commonly used and ultimately on the mission planner's choice (Fig. 2).

The rest of this chapter is organized as follows. First the technical characteristics and history of the main launchers employed by CubeSats are presented, grouping them into families of launch vehicles. Then the most important points to be considered when selecting a launch vehicle for a CubeSat mission are discussed.

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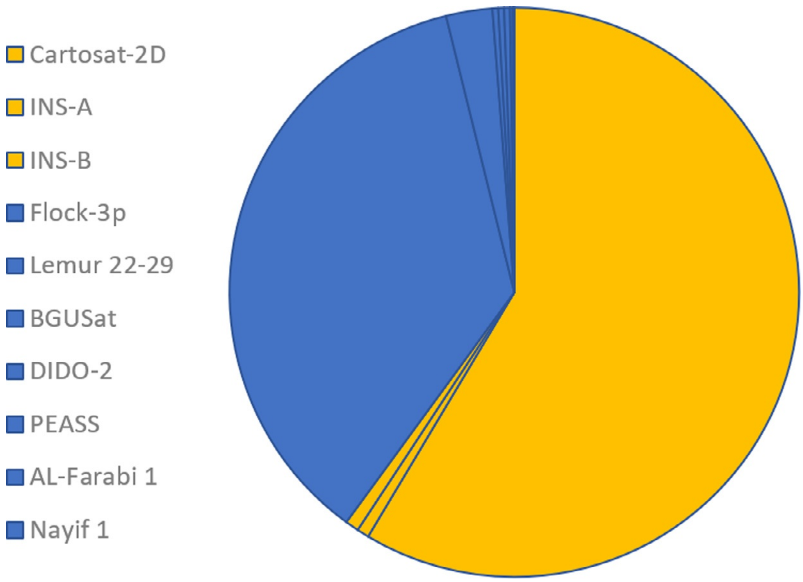

Fig. 1 PSLV-XL C37 Launch Manifest mass distribution. CubeSats' mass is shown in *blue* (*dark gray* in print version), and conventional satellite's is shown in *yellow* (*light gray* in print version).

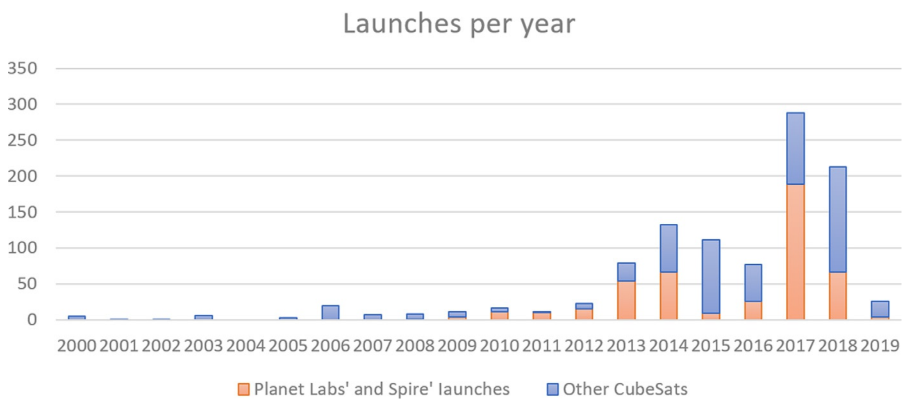

Fig. 2 Historical launchers of CubeSats from 2000 to March 2019.

2 Launch vehicle families

Launch vehicles normally belong to large families as their designers seek to provide optimized performance for several different payload scales, capacities and achievable orbits. Some liberties were taken when considering the families. For example, Minotaur and Antares were consolidated as a single family as they are both manufactured and

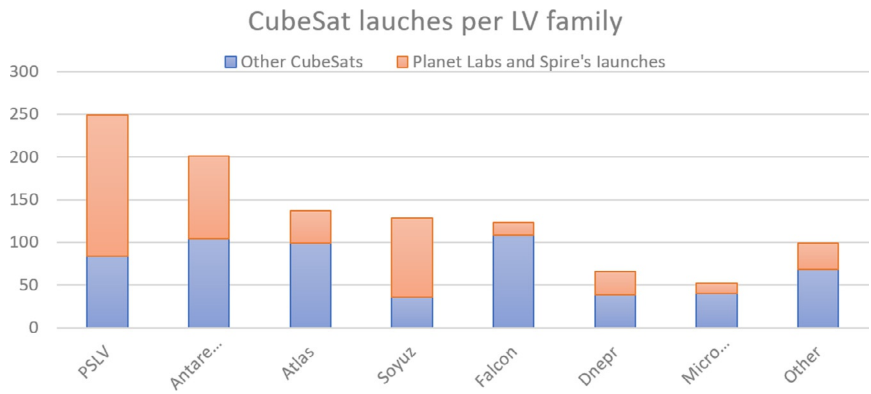

Fig. 3 Most frequently used launch vehicles for CubeSats from 2000 to 03/2019.

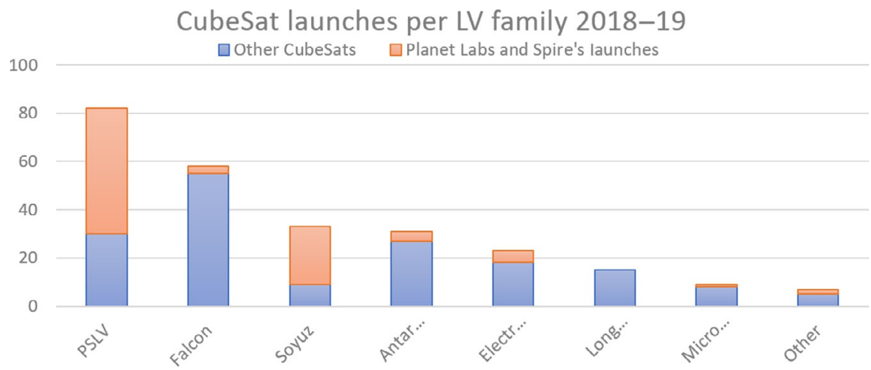

Fig. 4 Most frequently used launch vehicles for CubeSats from 2018 to 03/2019.

marketed by Northrop Grumman. The typical CubeSat launcher differs from the launch vehicles used on larger satellites, and some very successful vehicles such as Ariane-5 have never launched a CubeSat. The annual number of CubeSats has grown exponentially over the two decades of the standard's existence, and the launch vehicles employed by them have changed accordingly. Launchers like Soyuz that rarely transported CubeSats have become extremely relevant. Popular CubeSat carriers of the past, like Dnepr, have ceased operations. The landscape of CubeSat launches is set to change again with the consolidation and expansion of micro launch vehicles (Figs. 3 and 4).

2.1 Antares/Minotaur

The Antares/Minotaur launch vehicle family combines the launches by Orbital ATK's (currently Northrop Grumman) Minotaur, Taurus, and Antares rockets. The original vehicles Minotaur, Taurus, and Pegasus were all solid propellant stacks with motors developed by Thiokol and Alliant Techsystems [6]. Following a series of corporate mergers and acquisitions, Northrop Grumman consolidated control over the major subsystems of those vehicles.

Table 1 Antares 2 information.

Antares	Propellant	Engine/ motor	Thrust	Max G long.	Max G lateral
First stage	LOX/RP1	RD-180	1.92 MN	−1.0, +8.0	±1.5
Second stage	Composite	Castor XL	557 kN		
Payload to ISS orbit	6000–6500 kg (Cygnus capsule)				
Payload at 600 km SSO	3600 kg				
Commercial authority	Northrop Grumman				
Primary launch broker	NanoRacks				
Launch cost [10]	80–85 million USD, 12,880–12,900 USD/kg				
Principal launch site	Wallops Flight Facility				
Most notable flight for CubeSats	CRS Orb-2 08/13/2013, 32 CubeSats launched to ISS				
Greatest strength for mission planner	Frequent low-inclination low-altitude opportunities for CubeSats, direct deployment of release from the ISS				

The Antares/Minotaur family is the second most common launcher for CubeSats (second only to the PSLV family). This is mainly due to NanoRacks' ISS CubeSat Deployment and External Cygnus Deployment with 106 CubeSats deployed from ISS [7] and 30 deployed directly by the Cygnus capsule, which has typically employed Orbital ATK's launchers [7].

Minotaur-C is currently operational with a payload of 1590 kg to low Earth orbit [7]. It operates from four launch centers: US Air Force Western Range (WR) and Eastern Range (ER), NASA's Wallops Flight Facility (WFF), Pacific Spaceport Complex-Alaska (PSCA), and Reagan Test Site (RTS). Because of its many launch sites, Minotaur-C is capable of both polar orbits and low-inclination orbits (28.5 degrees), although commercial launches are only performed from WFF [8]. However, it is unlikely that lower inclinations are available to nongovernmental CubeSat launches as WFF does not serve this type of orbit. The Antares launch vehicle was developed to launch the Cygnus Spacecraft to the International Space Station under the COTS and CRS programs, although it can launch other dedicated satellites or CubeSat clusters. The Cygnus/ISS project has proven to be a reliable and consistent alternative for low-inclination and low-altitude CubeSat launches. Antares only launches from WFF.

Antares' first stage was developed by KB Yuzhnoye and is equipped with a RD-181 produced by NPO Energomash. Antares second stage was developed by Orbital ATK based on the Solid Rocket Motor CASTOR 30XL [9] (Table 1).

2.2 Dnepr

The Dnepr is a launch vehicle derived from the repurposing of 150 soviet R-36 ICMBs to serve as space launch platforms. The repurposing and later commercial exploitation of the missiles were tasked to ISC Kosmotras, which still runs the program [11]. Dnepr performed 22 space launches with a unique failure on July 26, 2006, causing the loss

Table 2 Dnepr information.

Dnepr	Propellant	Engine	Thrust	Max G long.	Max G lateral
First stage	UDMH/NTO	RD-264	4.52 MN	7.5 ± 0.5	± 0.8
Second stage	UDMH/NTO	RD-0255	755 kN	7.8 ± 0.5	0.5 ± 0.5
Third stage	UDMH/NTO	RD-864	20.2 kN	-0.3	0.25
Payload to ISS orbit	3000 kg				
Payload to 98° polar at 600 km	1200 kg				
Commercial authority	ISC Kosmotras				
Primary launch broker	Multiple launch brokers				
Principal launch site	Baikonur				
Launch cost [10]	29 million USD, 9063.00 USD/kg				
Most notable flight for CubeSats	19th flight 37 satellites deployed from 26 different organizations				
Greatest strength for mission planner	Talent on planning and managing large and diverse cluster launches				

of 14 CubeSats. Following tensions between the Russian and Ukrainian governments, no Dnepr launch has occurred since 2015.

Kosmotras ISC has shown an exceptional talent for organizing complex and diverse nanosatellite launches, with Dnepr's 19th launch holding the title of most satellites in a single launch until 2017 when PSLV's C37 launch surpassed it. Despite the impressive number of 37 satellites launched simultaneously, the most impressive figure of this launch was the participation of 26 independent satellite operators ranging from universities to national space agencies [7] (Table 2).

2.3 PSLV

The Polar Satellite Launch Vehicle (PSLV) is the Indian Space Agency's main low Earth orbit launcher. PSLV was initially developed to support the Indian remote sensing satellite program. PSLV has performed 48 missions [6, 12] with two failures and has performed launches ranging from low Earth orbit to geostationary and lunar orbits.

The PSLV stage configuration is composed of both solid and liquid propellant stages: with the first stage PS1 composed of an S139 and six to nine S12 strap on boosters, the second stage PS2 powered by the hypergolic engine Vilkas, the third stage PS3 with solid propellant motor, and the fourth stage PS4 equipped with a pressured-fed hypergolic engine [6].

PSLV currently holds the record for the most satellites launched on a single mission with launch C37 having carried 104 spacecraft of which 101 were CubeSats [4]. PSVL CubeSat payloads are primarily brokered by ISIS [7]. PSLV has currently only performed commercially Sun-synchronous launches for CubeSats with altitudes ranging from 485 to 780 km [2] (Table 3).

Table 3 PSLV-XL information.

PSLV-XL	Propellant	Engine	Thrust	Max G long.	Max G lateral
Boosters	Composite	$6 \times$ S12	6×719 kN	5 ± 1	± 0.5
First stage	Composite	S138	4.84 MN	5 ± 1	± 0.5
Second stage	UH25/NTO	Vilkas	803 kN	4.5 ± 0.2	± 0.6
Third stage	Composite	P37.5	240 kN	6.2 ± 0.2	± 0.5
Forth stage	MMH/MON-3	$2 \times$ L-2-5	14.6 kN	1 ± 0.2	± 0.5
Payload to ISS orbit	5950 kg				
Payload to 600-km SSO	4550 kg				
Commercial authority	Antrix				
Primary launch broker	ISIS				
Principal launch site					
Launch cost [10]	21–31 million USD, 6642.00–9538.00 USD/kg				
Most notable flight for CubeSats	C37 holds the record for largest number of satellites deployed on single launch 104 (101 were CubeSats)				
Greatest strength for mission planner	Low launch cost and frequent access to higher polar orbits for CubeSats				

2.4 Soyuz-2

Soyuz-2 and its variants are the current version of the oldest launch vehicle family in history, dating back to Sputnik's launch in 1957. The Soyuz rocket is the most utilized launch vehicle in history, with more than 1700 launches [6]. Historically, Soyuz has launched very few CubeSats, having launched only 18 spacecraft until June 2017. From then on, this changed, and Soyuz has since launched 108 CubeSats on four different missions [7].

Soyuz's main booster is composed of three stages: the first stage, the boosters, is powered by four RD-107A engines; the second stage is powered by a single RD-108A; and the third stage by one RD-0124 [13]. Soyuz also employs the Fregat-M upper stage for launche that need to achieve higher altitudes [14].

CubeSat launches of Soyuz are currently mostly brokered by EXOLaunch and ISIS [7]. Soyuz usually launches CubeSats to high-inclination orbits. Soyuz launches from Baikonur, Plesetsk, Kourou, and Vostochny [13, 14], although CubeSats have never been launched from Kourou (Table 4).

2.5 Falcon 9

Falcon 9 is SpaceX's main launch vehicle for geostationary, low Earth orbit, and ISS resupply. Falcon 9 and Falcon Heavy are currently the only partially reusable launch systems in operation. SpaceX successfully lowered the space launch cost by at least two-thirds [10] and is currently developing an interplanetary super heavy launch system and a manned capsule to service the ISS. In 2019 SpaceX announced the first structured rideshare program to allow booking of secondary payloads independently

Table 4 Soyuz-2-1a information.

Soyuz-2-1a	Propellant	Engine	Thrust	Max G long.	Max G lateral
First stage (boosters)	RP1/LOX	RD-107A	4×838.5 kN	4.3 ± 0.7	± 1.8
Second stage	RP1/LOX	RD-108A	792.5 kN	2.6 ± 1.2	± 0.8
Third stage	RP1/LOX	RD-0124	297.9 kN	2.2 ± 1.5	± 0.3
Forth stage (Fregat)	UDMH/NTO	S5.92	19.6 kN	N/A	N/A
Payload to ISS orbit	5950 kg				
Payload to 98° polar at 600 km	4550 kg				
Commercial authority	Starsem (Soyuz-2-1a)				
Primary launch broker	EXOLaunch				
Principal launch site	Baikonur				
Launch cost [10]	80 million USD, 16,495.00 USD/kg				
Most notable flight for CubeSats	67 CubeSats launched on 07/14/2017				
Greatest strength for mission planner	Possible customized deployment of CubeSat constellations due to Fregat high maneuver capacity				

from the primary; the program also allows for rescheduling of a launch without cost in the event of mission delay [15].

Falcon 9 utilizes RP-1 and liquid oxygen on all its stages. The first stage is equipped with nine Merlin 1D engines, and the second stage is equipped with a single MVac, the Vacuum variant of the Merlin engine [10]. The current version launcher's first stage is also equipped with fast reusability systems: landing legs, titanium grid fins, and nonpyrotechnical actuators.

Falcon 9 operates from Cape Canaveral Air Force Station, Kennedy Space Center, and Vandenberg Air Force Base. SpaceX is currently building a private launch center in Texas. Falcon 9 has launched CubeSats to orbits ranging in inclination from 34.5 to 98 degrees and on altitudes from 245 to 580 km [6]. Falcon 9 CubeSat launches are brokered by several companies: Spaceflight, Tyvak, NanoRacks, and ISIS (Table 5).

2.6 Atlas V

Atlas V is the current version of one of the oldest American launch vehicle families with the first launch having occurred in 1959 [6]. Atlas V is manufactured and operated by United Launch Alliance and together with Delta IV handles most US military space launches as part of the National Security Space Launch (NSSL), formerly Evolved, Expendible Launch Vehicle (EELV) [18]. Atlas V has also launched many American interplanetary missions including the only interplanetary CubeSats, MarCO A and B [2].

Atlas V is a two-stage launch vehicle composed of the Common Core Booster (CCB) as its first stage and a Centaur upper stage as its second. CCB is powered

Table 5 Falcon 9 information.

Falcon 9 Block 5	Propellant	Engine	Thrust	Max G long.	Max G lateral
First stage	RP1/LOX	Merlin-1D	$9\times$ 854 kN	6 (8.5[a])	2 (3[a])
Second stage	RP1/LOX	MVac	981 kN		
Payload to ISS orbit	8700 kg[b]				
Payload to 98° polar at 600 km	7500 kg[b]				
Commercial authority	SpaceX				
Primary launch broker	Multiple launch brokers				
Principal launch site	Kennedy Space Center				
Launch cost [4]	60.1 million USD, 2864.00 USD/kg				
Most notable flight for CubeSats	Spaceflight SSO-A Sherpa Launch				
Greatest strength for mission planner	Cheapest launcher per kilogram, wide range of inclinations, and SmallSat program could allow better constellation deployments				

[a] Lateral and longitudinal loading for payloads of less than 1814 kg.
[b] Falcon's payload performance is no longer publicly available [16]; the figures presented are from Falcon 9 v1.0 [17].

by a single RD-180 dual-chamber liquid propellant engine [19]. The Centaur is powered by a RL-10 cryogenic engine [19]. Atlas V's performance can be augmented by several configurations of strap-on boosters: from parallel CCB on the HLV variant to solid rocket motors on the 400 and 600 series [19].

Atlas V launchers from both Vandenberg AFB (WR) and Cape Canaveral AFS (ER) provide opportunities for both low-inclination (geostationary and ISS) and high-inclination launches [19]. Following 2014 Antares Accident Orbital ATK has contracted ULA to launch Cygnus. Atlas V CubeSat launches are usually brokered by Tyvak except for the three Cygnus launches, which were majorly brokered by NanoRacks [7]. Atlas V has deployed CubeSats on both ISS and polar orbits on altitudes ranging from 400 km to interplanetary [7] (Table 6).

2.7 *Micro satellite launchers*

Although micro satellite launchers have been in operation since 2006, with the M-5's maiden flight, until recently, none of those vehicles had operated constantly, having performed only proof-of-concept flights or tests. This changed in 2017 with frequent launches of Long March 11, KZ-1A, Epsilon, Electron, and SS-520. From 2018 to 2019, Electron became the fifth most used launcher for CubeSats, and overall micro launchers contributed 12% of the total CubeSat launched on the period [2]. The popularization of micro launchers has opened an unprecedented opportunity for CubeSat mission planners allowing them to design systems where their spacecraft are the primary payloads and where CubeSat orbits can be designed without the burden imposed by ridesharing.

Table 6 Atlas V-401 information.

Atlas V-401	Propellant	Engine	Thrust	Max G long.	Max G lateral
First stage	RP1/LOX	Merlin-1D	3.82 MN	5	2
Second stage	RP1/LOX	RL-10-A	99.2 kN		
Payload to 28.5° at 600 km	9224 kg				
Payload to SSO polar at 600 km	7434 kg				
Commercial authority	ULA				
Primary launch broker	Tyvak				
Principal launch site	Eastern Range and Vandenberg				
Launch cost [4]	131–179 million USD, 9514.00–12,903.00 USD/kg				
Most notable flight for CubeSats	MarCO A and B, the first interplanetary CubeSats				
Greatest strength for mission planner	Strong utilization by US governmental launches				

Table 7 Micro launch vehicle launches in 2017–19.

Micro launchers	Launches	Fails	Note
Electron	8	1	Private enterprise, has transported CubeSats
CZ-11	5	0	Chinese, has transported CubeSats
OS-M1	1	1	Chinese Private Enterprise, has transported CubeSats
SS-520	2	1	Has transported CubeSats
Epsilon	2	0	Has transported CubeSats
LandSpace-1	1	1	Chinese Private Enterprise
KZ-1A	5	0	Chinese Private Enterprise
Hyperbola-1	1	0	Chinese Private Enterprise
Jielong-1	1	0	Chinese Private Enterprise

From 2018 to 2019, five different micro launch vehicles transported CubeSats: Electron, OS-M1, Long March 11, Epsilon, and SS-520. Although one SS-520 and one OS-M1 launches failed, the micro launcher trend is persisting (Table 7).

Between 2017 and 2019, nine micro launch vehicles have operated—six of them being Chinese, two Japanese, and one from New Zealand. Six of the nine micro launch vehicles were developed or are operated by private companies. Yet, it is worth noticing that some of the companies are spin-offs of public entities such as China Rocket (Jielong-1's operator), which is a subsidiary of the China Academy of Launch Vehicle Technology (CALT).

The recent increase of commercially available micro launch vehicles opens opportunities for the CubeSat mission planner allowing for freedom both of schedule and orbital design. The micro satellite launchers' payloads vary from the 4 kg on SS-520

Table 8 Electron information.

Electron	Propellant	Engine	Thrust	Max G long.	Max G lateral
First stage (boosters)	RP1/LOX	Rutherford	9×16 kN	±6	±0.7
Second stage	RP1/LOX	Rutherford Vac	22.1 kN		
Third stage (optional)	Monopropellant[a]	Currie	120 N		
Payload to SSO polar at 600 km	142 kg				
Commercial authority	Rocket Labs				
Primary launch broker	Rocket Labs				
Principal launch site	Rocket Lab Launch Complex 1				
Launch cost [4]	4.9 million USD, 32,667 USD/kg				
Most notable flight for CubeSats	14 CubeSats successfully launched by a commercial micro satellite launcher				
Greatest strength for mission planner	Small CubeSat constellation could be deployed as main payload on predesigned orbit				

[a] Green monopropellant.

to 500 kg on Epsilon (500-km SSO). The available range of payloads allow for single CubeSat deployments, although economically ineffective, to consistent cluster launching and constellation building (Table 8).

3 Launch vehicle selection: Alternative solutions

Launch vehicle selection is an often overlooked concern on CubeSat missions, mainly due to containerization and lack of influence on the launch provider. In a traditional piggyback launch, CubeSat operators can neither choose nor influence the final orbit, and they are released on the same orbit as the main payload. An exception is represented by launchers with maneuvering final stages, but even in this case, CubeSats do not have control over the destination orbit, since the final-stage maneuvers are performed to protect the main payload. Several missions cannot be launched as piggyback launches, since the typically available orbits (Sun-synchronous orbits and those obtained with the launches from the ISS) do not match the ones needed to accomplish the mission goals. Examples of this are highly elliptical orbits for radiation belt studies, equatorial orbits for high availability communications, and halo orbits for astronomy, among others.

In the case of CubeSat missions, the only alternative usually available for the mission planner is to adapt the mission design to the available launch opportunities or to wait until a bigger mission is scheduled for launch on the same desired orbit. There are several cases of CubeSat missions that have learned to adapt to the available launch opportunities: Planets Labs' agile aerospace method recommends launching as often as possible and on as many launchers as possible [20] to avoid single point failures on constellation building. Other missions, like Lunar IceCube [21], are completely dependent on NASA's SLS program schedule.

The primary concerns related to launch vehicle selection for CubeSat missions are the availability of the intended orbit, the frequency of launches, and the types of CubeSat deployers that serve the chosen launch vehicle. When it comes to building a constellation, the situation becomes more challenging, because the mission planner must secure not only a single adequate opportunity but also a constant stream of launches.

As seen in the previous section, a recent trend is that of micro launch vehicles, such as Electron and CZ-11. The smaller capacities of micro launch vehicles allow for cluster launches to be easily organized, since a smaller number of CubeSats are required to fully occupy the launch vehicle. Another advantage of micro launch vehicles is the possibility of reaching tailored orbits: a CubeSat customer can contract the entire payload capacity selecting a specific orbit, something previously impossible for CubeSats [22]. This freedom of design allows for great improvements, especially on CubeSat communication constellations [23].

SpaceX's SmallSat and other similar programs under development might also improve the available options for CubeSat launches, in particular for constellation building. Nevertheless, they are less disruptive than micro launchers for the users, since they are still limited to the orbital destinations provided by SpaceX. The SmallSat program provides launches for 150- and 300-kg payloads to be contracted independently of the primary payload [15]. It will give more flexibility since the first scheduling and possible postponement of the main payload will not affect the secondary one and vice versa. The 150-kg payload can also accommodate a small satellite equipped with CubeSat deployers, similarly to GAUSS's UniSat solution [24]. Finally, it is also possible to design a powered space tug to be launched on a SmallSat slot and then perform maneuvers to correctly deploy a constellation of CubeSats on a tailored arrangement [25].

The selection of an available launcher for a CubeSat mission is a complex process. The major drivers and decision points are summarized in the flowchart in Fig. 5. The first point to be considered when selecting a launcher is the desired orbit. As previously stated, there are numerous opportunities for low Earth orbit launches but frequent interplanetary launch opportunities do not yet exist. The alternatives for interplanetary CubeSat missions are either redesigning the mission or waiting for a larger interplanetary mission to piggyback on. One redesign alternative is to fly LEO Proxy missions where not all scientific objectives are met, but the research can move forward. Lunar missions still have the alternative to launch to geostationary transfer orbit (GTO), raise the orbit, and be captured by the Moon; nevertheless, this requires either high ΔV from onboard propulsion systems or advanced low-ΔV trajectories, described in Chapter 1. If no redesign is possible or desirable, the remaining alternative is to wait for an interplanetary mission.

CubeSat missions that require orbits different from the typical ISS or polar orbits have very few launch alternatives at their disposal, nowadays. The situation for these missions is similar to the one experienced by the interplanetary CubeSats. Fortunately, for LEO CubeSats, there is the possibility of launching on a micro launch vehicle. The most frequently launched micro rockets are Electron, CZ-11, and KZ-1A.

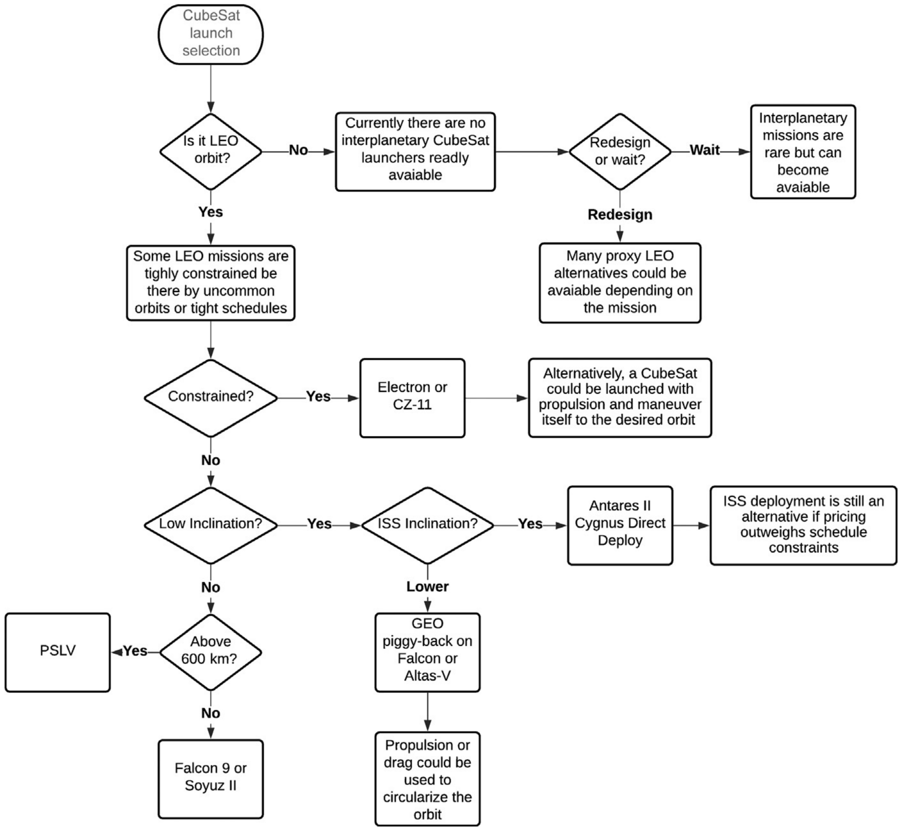

Fig. 5 Flowchart of CubeSat launch vehicle selection.

This alternative is expensive and may not be available to most missions. The SS-520 launcher can deliver a single CubeSat to LEO and might be a viable opportunity for constrained missions; notwithstanding the SS-520 is not planned for serialized production.

If the mission is not constrained by tight schedules or unconventional orbits, more alternatives are possible. Low-inclination missions can be launched from the ISS, but the on-station storage time might be an issue to be considered. The recommended alternative is Cygnus' direct deployment where the payload is inserted into' a near ISS orbit without the storage period. ISS deployment is still interesting, especially to educationally oriented missions since it adds the bonus of interacting with the astronauts and the station itself. ISS launches are described in details in Chapter 24.

There are very few alternatives for rides to low-inclination or equatorial orbits. CubeSat missions requiring these kinds of orbits might have only two options: micro launch vehicles or orbital maneuvers. However, no micro launchers are currently operated from equatorial launch sites and consequently will need to perform highly energetic maneuvers to change the orbital inclination. Henceforth the only remaining option for a CubeSat to reach equatorial LEO might be to launch to GTO and reduce the orbital apogee to reach LEO. Both passive [26] and active [27] alternatives have been proposed, but these alternatives' have never been tried and would add considerable risk to a mission. Both Atlas V and Falcon 9, however, have frequent opportunities for GTO CubeSat launches.

If the mission targets a polar orbit or Sun-synchronous orbit, it has numerous available alternatives, nowadays. For higher altitudes, PSLV is the recommended launcher for both launch frequency and pricing. For lower-altitude SSO orbits (below 600 km), the recommended vehicles are Soyuz and Falcon 9 also due to the frequency of launches and pricing.

4 Conclusions

The CubeSat launcher market is changing rapidly, and new opportunities and challenges will no doubt appear in the future. The consolidation of smaller launch vehicles and new launch arrangements is an important factor to be considered by the mission planner. Identifying a launching strategy and a proper launcher is as important to a CubeSat mission as securing access to space or funding. The process of selecting a launcher is a multidisciplinary process and could be a major mission driver.

References

[1] CubeSat Design Specification Rev. 13 The CubeSat Program, Cal Poly SLO, 2014.

[2] M. A. Swartwout, University of Saint Lois's CubeSat Database, Accessed 15 August 2019.

[3] M. Swartwout, Secondary spacecraft in 2016: why some succeed (and too many do not), in: 2016 IEEE Aerospace Conference, IEEE, 2016, March, pp. 1–13.

[4] Indian Space Research Organization, Publication and Public Relations ISRO Headquarters, Antariksh Bhavan, PSLV-C37 Cartosat-2 Launch Brochure.

[5] M. Tugnoli, M. Sarret, M. Aliberti, European Access to Space: Business and Policy Perspectives on Micro Launchers, Springer Briefs in Applied Sciences and Technology, 2019, ISBN 978-3-319-78960-6. https://doi.org/10.1007/978-3-319-78960-6.

[6] S.J. Isarowitz, J.B. Hopkins, J.P. Hopkins Jr., "International Reference Guide to Space Launch Systems". fourth ed., American Institute of Aeronautics and Astronautics, Reston, VA, 2004. https://doi.org/10.2514/4.475917 USA.

[7] Erik Kulu, NanoSat Database, https://www.nanosats.eu/database, Accessed 15 August 2019.

[8] Minitour's User Guide Release 1.0, March 2002.

[9] Antares User's Guide Release 3.0, August 2018.

[10] Surplus Missile Motors, Sale Price Drives Potential Effects on DOD and Commercial Providers, United States Government Accountability Office, Washington, DC, August 2017.

[11] Space Launch System Dnepr User's Guide, Issue 2, November 2001, Surplus Missile Motors, Sale Price Drives Potential Effects on DOD.

[12] PSLV, User's Manual, ISRO, Issue 3, May 1996.

[13] Soyuz User's Manual, Issue 2 Revision 0, Arianespace, March 2012.

[14] Soyuz User's Manual, ST-GTD-SUM-01, Issue 3, Revision 0, Starsem, April 2001.

[15] SpaceX's SmallSat Program Webpage, https://www.spacex.com/SmallSat, Accessed 15 August 2019.
[16] Falcon User's Guide, January 2019.
[17] Falcon 9 Launch Vehicle Payload User's Guide, Rev 1, 2009.
[18] J.L. Buzzatto, The Transition of Space Launch—Heritage to EELV, Space, Long Beach, CA, 2003. 23–25 September 2003.
[19] V. Atlas, Launch Services User's Guide, March 2010.
[20] M. Safyan, Astronautics Smallsat launch strategy, Room Space J. 2 (9) (2016).
[21] P.E. Clark, et al., Lunar Ice Cube mission: determining lunar water dynamics with a first generation deep space CubeSat, in: 47th Lunar and Planetary Science Conference, 2016.
[22] Payload User's Guide, Version 4.0, December 2016.
[23] Nano/Microsatellite Market Forecast, ninth ed., SpaceWorks Enterprises, Inc. (SEI), 2019 Copyright 2018.
[24] C. Cappelletti, S. Battistini, F. Graziani, Small launch platforms for micro-satellites, Adv. Space Res. 62 (13) (2018) 3298–3304.
[25] J. Andrews, Spaceflight secondary payload system (SSPS) and SHERPA tug—a new business model for secondary and hosted payloads, in: 26th Annual AIAA/USU Conference on Small Satellites, 2012.
[26] C. Lücking, C. Colombo, C.R. McInnes, Mission and system design of a 3U CubeSat for passive GTO to LEO transfer, in: 63rd International Astronautical Congress, 2012.
[27] D.T. Schmuland, C. Carpenter, R.K. Masse, Mission applications of the MRS-142 CubeSat High-Impulse Adaptable Monopropellant Propulsion System (CHAMPS), in: 48th AISS/ASME/ASEE Joint Conference & Exhibit, 2012.